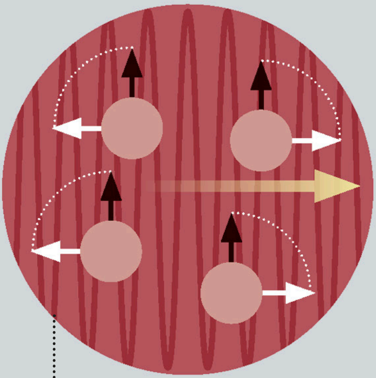
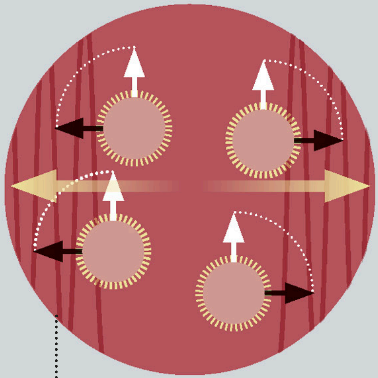


RADIO PULSE MOVEMENT

Radio wave pulses of a specific frequency are sent through the body, causing the protons to vibrate and exciting the protons into a different alignment (antiparallel).



RELEASE OF RADIO SIGNAL

When the radio pulses stop, the protons keep vibrating and gradually realign with the magnetic field. Their vibration releases measurable radio waves (energy) in the process.

LOOKING INSIDE

Magnetic resonance imaging (MRI) is a noninvasive medical imaging technique. The patient is placed in a powerful magnetic field, forcing the magnetic moments (see p.67) of protons in the body to align. Protons make up the nuclei of atoms of hydrogen, one of the most abundant elements in the body. When a current is applied, the protons become excited and spin (or vibrate) out of position. Measuring the frequency of the vibrations of the protons after the current is switched off allows different tissues to be distinguished.